

Dimension Reduction Analysis: pbmc

Generated report #1

Seed: 0 | Plan revision: 1

Dataset profile & preprocessing

Peripheral blood mononuclear cells (PBMCs) from a healthy donor, profiled with 10x Genomics single-cell RNA sequencing (scRNA-seq). This is the standard "pbmc3k" dataset used throughout the Scanpy tutorials and widely used as a benchmark for single-cell analysis methods. Stored as the raw h5ad file from the Scanpy tutorials.

Samples: 2,700 Features: 32,738 Sparsity: 97.4%

Missing values: No

Preprocessing: normalize_total

profile.json shows 97.4% zeros and non-negative integer values in `[0, 419]`, and `DATA_DESCRIPTION.md` says these are raw UMI counts with total counts varying across cells (mean ~2,367 per cell, genes detected ranging 212 to 3,422). Rescaling each cell to a common total removes this sequencing-depth difference so it does not become the dominant axis of variation. `DATA_DESCRIPTION.md` states no normalization was applied upstream, so this is not double-normalizing. `target_sum` is left at the implementation default, the median per-cell total count; the exact constant only rescales every cell equally before `log1p` and does not change the method choice.

Preprocessing: log1p

`DATA_DESCRIPTION.md` warns the counts are heavy-tailed, with a few highly expressed genes dominating raw totals (max value 419 against a mean of 0.072 per profile.json). `log(1+x)` compresses that tail so a handful of genes do not dominate every distance-based method, while mapping the 97.4% zeros to zero.

Dataset profile & preprocessing (cont.)

Preprocessing: select_top_variance

n_features (32,738) is about 12x n_samples (2,700), and most genes are zero in nearly every cell (97.4% sparsity), so they add noise but little signal to distances. Keeping the 2,000 highest-variance genes (after normalize + log1p) makes Kernel PCA and Diffusion Maps' N x N distance computations cheaper and less noise-driven. Judgment call: this ranks by raw variance, not by mean-adjusted dispersion as scRNA-seq tools usually do, so some high-mean genes may be favored; the alternative (keeping all genes) was rejected because it would let tens of thousands of near-empty genes dilute the distances. No standardize step follows: per data-profiling/SKILL.md, standardizing on a comparable scRNA-seq set gave every low-expression noisy gene equal weight and changed fine neighborhoods substantially, with no labels here to show that as an improvement. Known limitation: no cell or gene QC filtering is available in this pipeline, although DATA_DESCRIPTION.md says the data are unfiltered (possible low-quality cells and doublets remain).

Methods and hyperparameters

pca (role: general_purpose, status: reused)

Hyperparameters: `n_components=50`

Required linear baseline. `n_components = min(50, floor(2700/2)=1350, 2000) = 50`. With 2,700 cells and 2,000 retained genes, 50 components is enough to summarize the main expression programs while staying far below the sample count; the sidecar reports how much variance they capture and how many components 90% would need. Also the source for the clustering used to color these unlabeled plots.

kernel_pca (role: general_purpose, status: reused)

Hyperparameters: `n_components=50, kernel=rbf, random_state=0`

Required PCA variant. `n_samples = 2,700` is under the ~5,000 threshold, so the 2,700 x 2,700 RBF kernel matrix is affordable, and Kernel PCA can capture nonlinear structure that linear PCA misses in the log-expression space. Sparse PCA was the alternative; at this sample size cost does not force it, and it targets interpretable (sparse) loadings rather than nonlinear structure. `n_components = 50` matches PCA for a like-for-like comparison. Judgment call: the RBF gamma is left at sklearn's default `1/n_features (1/2000)`; no data-driven bandwidth is available without labels, and the diffusion-map entry below uses a median-distance bandwidth, giving a second, differently-tuned kernel view.

diffusion_maps (role: general_purpose, status: reused)

Hyperparameters: `n_components=10, alpha=1.0`

Optional second nonlinear general-purpose method, affordable at 2,700 cells ($N \times N$ matrix). `DATA_DESCRIPTION.md` describes PBMCs, a mix of blood cell populations where continuous differentiation-like gradients as well as discrete groups are plausible, and diffusion maps are well established for scRNA-seq for exactly that kind of structure. 10 components, not 2, since its role is a quantitative embedding; the first two are plotted. epsilon defaults to the median pairwise squared distance (judgment call: no labels to tune it against).

Methods and hyperparameters (cont.)

umap (role: visualization_only, status: reused)

Hyperparameters: `n_components=2`, `n_neighbors=15`, `min_dist=0.1`, `random_state=0`

Visualization only. `n_samples = 2,700` falls in the 500 to 20,000 range where `n_neighbors = 15` is the heuristic; `min_dist = 0.1` is UMAP's own default. With no labels, this is the main picture for judging whether the PCA-derived clusters look like separated groups. Its embedding is never used for clustering or any other computation.

tsne (role: visualization_only, status: reused)

Hyperparameters: `n_components=2`, `perplexity=30`, `random_state=0`

Added for a specific reason: the dataset has no labels, and CLAUDE.md requires a second setting before trusting any structural claim from a manifold method. Only one UMAP artifact per method name is supported, so t-SNE (a different objective on the same preprocessed data) serves as the cross-check: a group is only described as separated in the report if it appears in both UMAP and t-SNE. Perplexity 30 is well under $n_samples/3 = 900$. Visualization only, never used for any computation.

Methods and hyperparameters (cont.)

Data-derived clusters (k-means on pca; colors only)

k=4 was chosen as the value with the highest silhouette among k=4 to 10 (k=4: 0.289, k=5: 0.256, k=6: 0.204, k=7: 0.206, k=8: 0.142, k=9: 0.129, k=10: 0.102). Cluster sizes: 1,330, 685, 348, 337. These clusters only color the plots of an unlabeled dataset; they are a description of the embedding, not a finding, and no biological or domain meaning is attached to them.

profile.json has `has_labels: false`, so without clusters every figure would be a single-color density plot. k-means runs on the 50-component PCA embedding (general_purpose; UMAP/t-SNE are never a clustering source). Judgment call, revised twice. First revision: the default scan [2, 10] picked k=2, the lower bound, but PCA/UMAP/t-SNE all showed the larger of those two clusters splitting into a big group plus a separate compact group of ~350 cells, so two colors hid visible structure; raising the floor to 3 gave k=3 (silhouette 0.304). Second revision, made after checking directly rather than trusting the UMAP picture: UMAP's shapes are never ground truth on their own (CLAUDE.md guard rails), so before raising k again this was verified against the general_purpose PCA embedding itself, not the UMAP layout. A density-based grouping of the UMAP layout (DBSCAN, used only to name candidate point sets for this check, never as a clustering result) isolates that same ~350-cell island; cross-checking those cell indices against k-means on PCA shows 349 of 351 land in the same PCA cluster as the main body at k=3, but 345 of 351 form their own PCA cluster at k=4 (silhouette 0.289, barely below k=3's 0.304). So the group is genuinely separable in the general-purpose embedding, not just an artifact of UMAP's layout; k=3 was merging it purely because silhouette favors coarse splits. Alternatives: keep `k_range` at [3, 10] (known from this check to still hide the compact group), or fix k=4 outright (rejected: no basis to also rule out 5-10 without the same scan). Raising the floor to 4 keeps the choice data-driven by silhouette while excluding the two floors already shown to under-cluster. Clusters are for coloring only and are not named as cell types.

Quantitative metrics

method	trustworthiness	variance explained (PCs used)	PCs for 90% variance
pca	0.9344	0.2904	>200
kernel_pca	0.9199		
diffusion_maps	0.8466		
umap	0.7740		
tsne	0.8023		

How to read the metrics

Trustworthiness (`sklearn.manifold.trustworthiness`, 10 neighbours)

For each sample, takes its 10 nearest neighbours in the embedding and penalizes any that were not close neighbours in the original preprocessed data, weighting each penalty by how far down the original neighbour ranking that point sat. Ranges 0 to 1; 1 means the embedding introduces no false neighbours. It measures local neighbourhood fidelity only, not preservation of global distances. Computed against the data after the plan's preprocessing steps; on a random 5,000-sample subset when the dataset is larger than that (see Limitations).

Variance explained (PCs used)

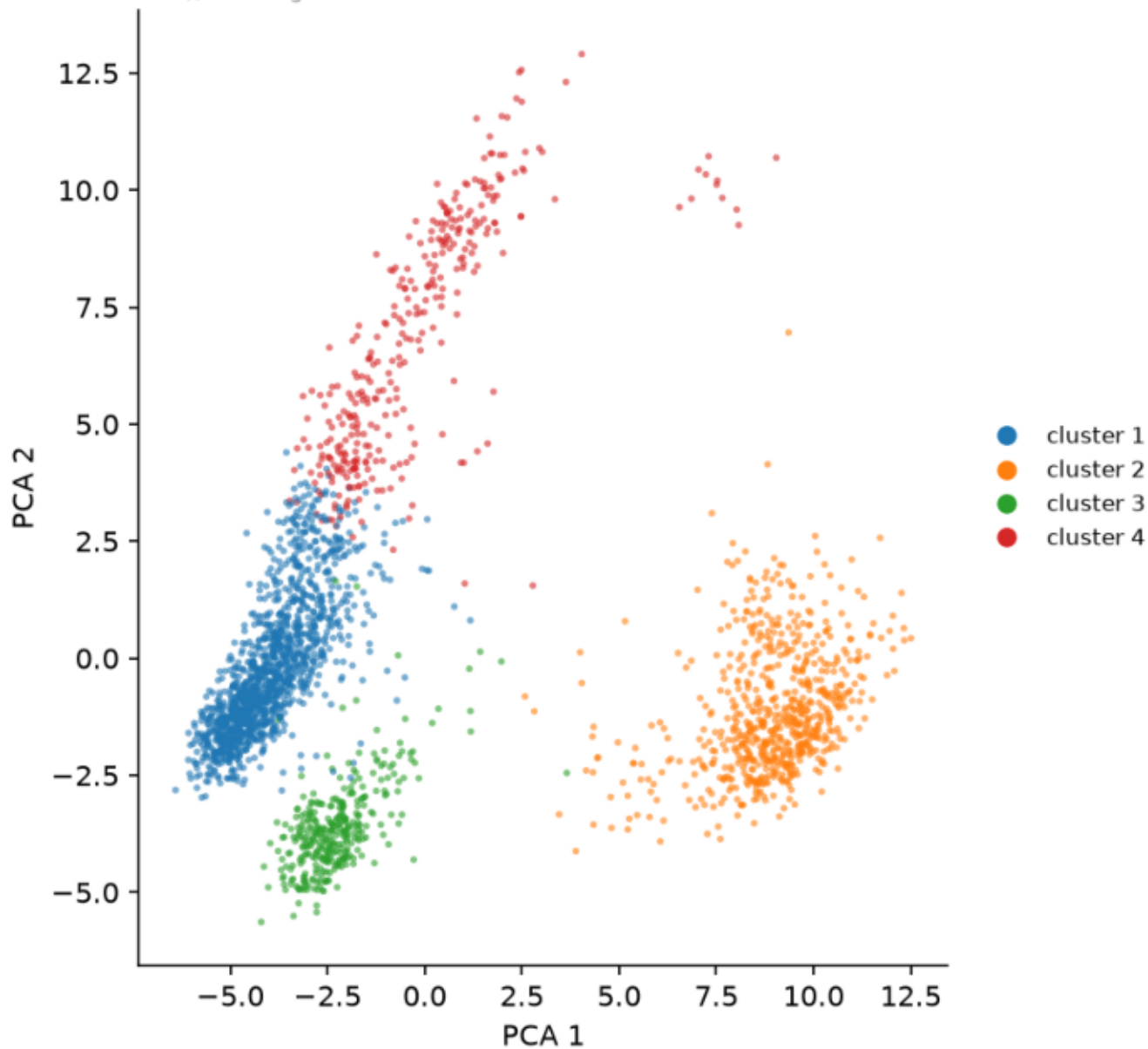
The fraction of the total variance in the preprocessed data captured by the principal components actually kept in the embedding: the sum of those components' explained-variance ratios (each component's variance divided by the total). Ranges 0 to 1. PCA only; blank for methods it does not apply to.

PCs for 90% variance

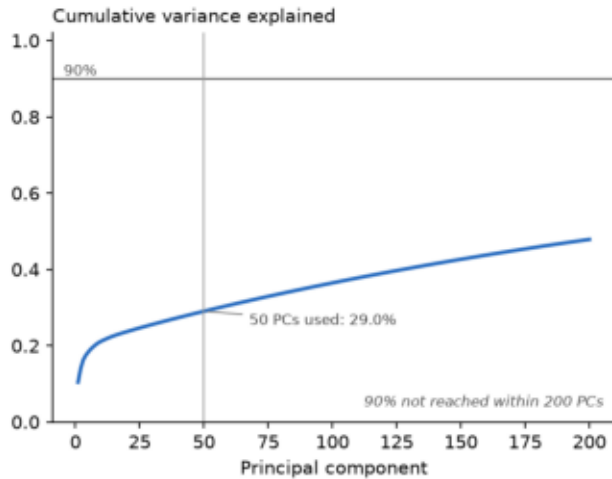
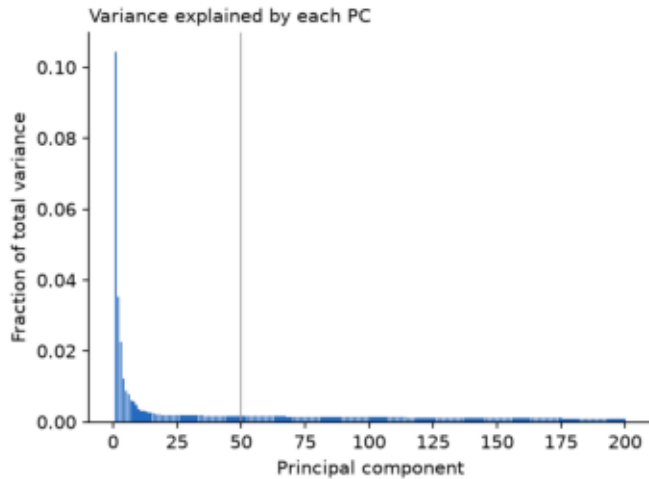
The smallest number of principal components whose cumulative explained variance reaches 90%, read off a diagnostic PCA fit of up to $\min(4 \times n_{\text{components}}, 200)$ components. A leading '>' means 90% was not reached within that fit, so the true number is larger than shown. The scree plots show the full curve.

pbmc: pca

n=2,700, colored by data-derived clusters (k-means on pca, k=4), showing dims 1-2 of 50

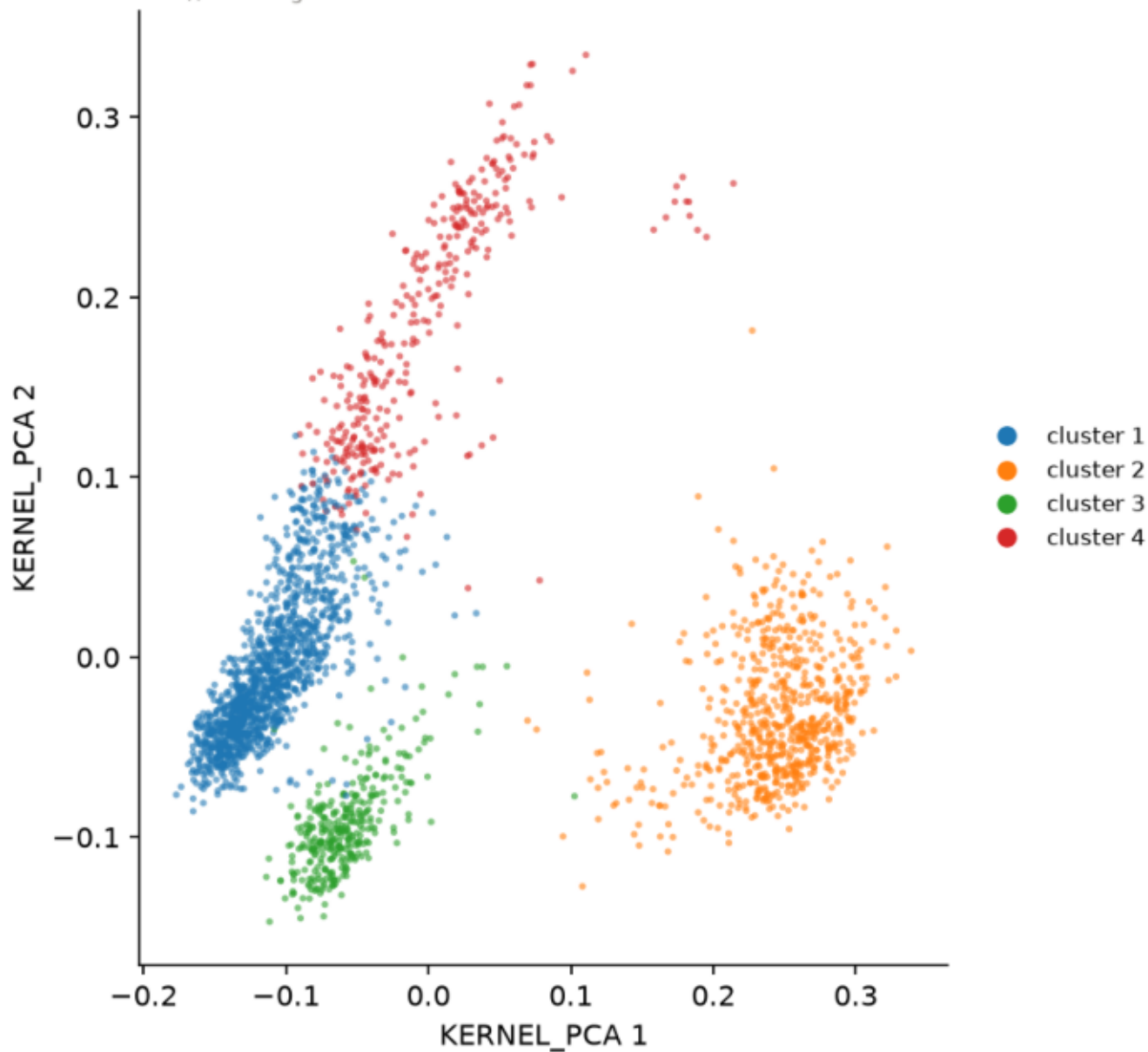


pbmc: pca scree plot



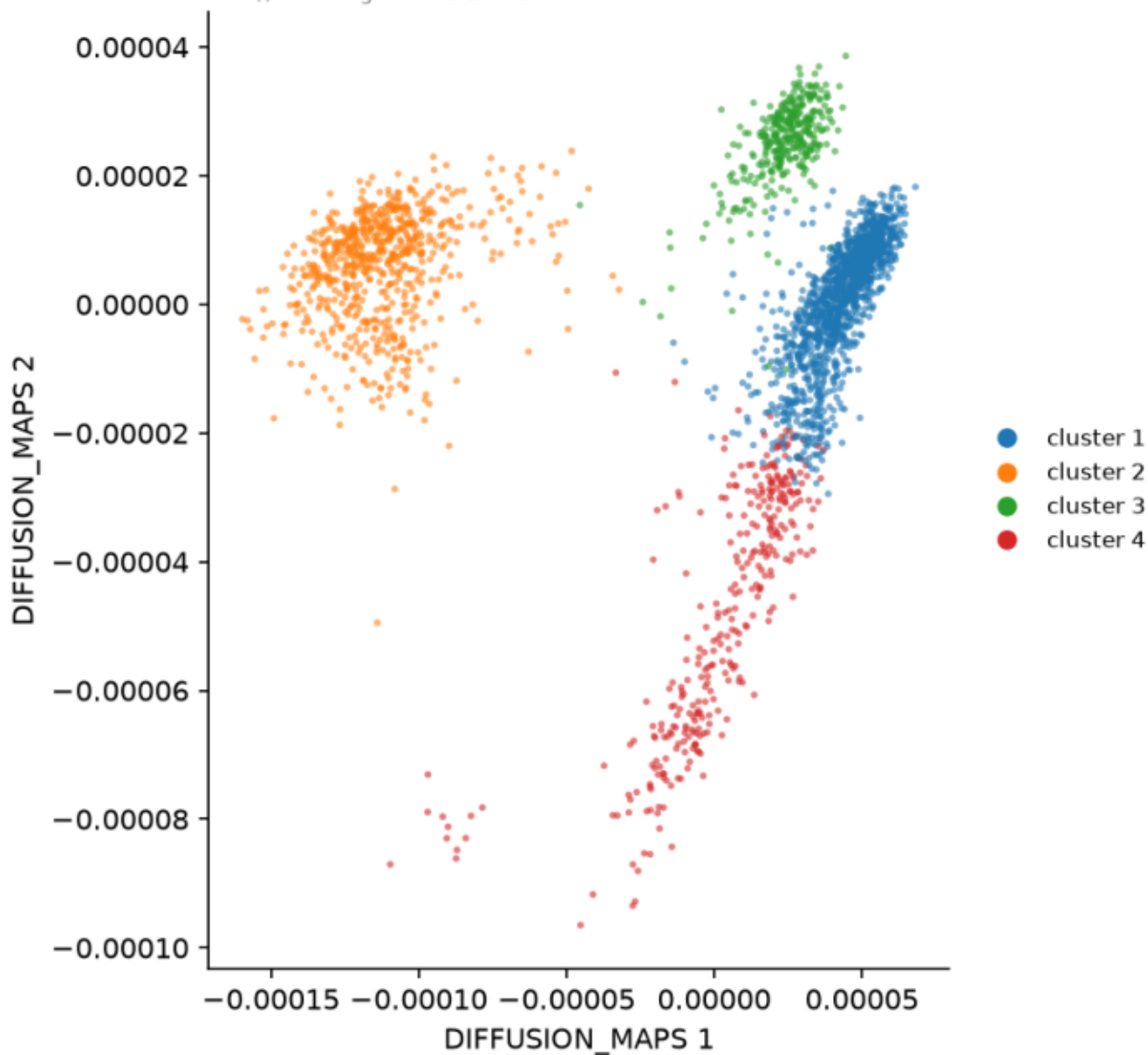
pbmc: kernel_pca

n=2,700, colored by data-derived clusters (k-means on pca, k=4), showing dims 1-2 of 50



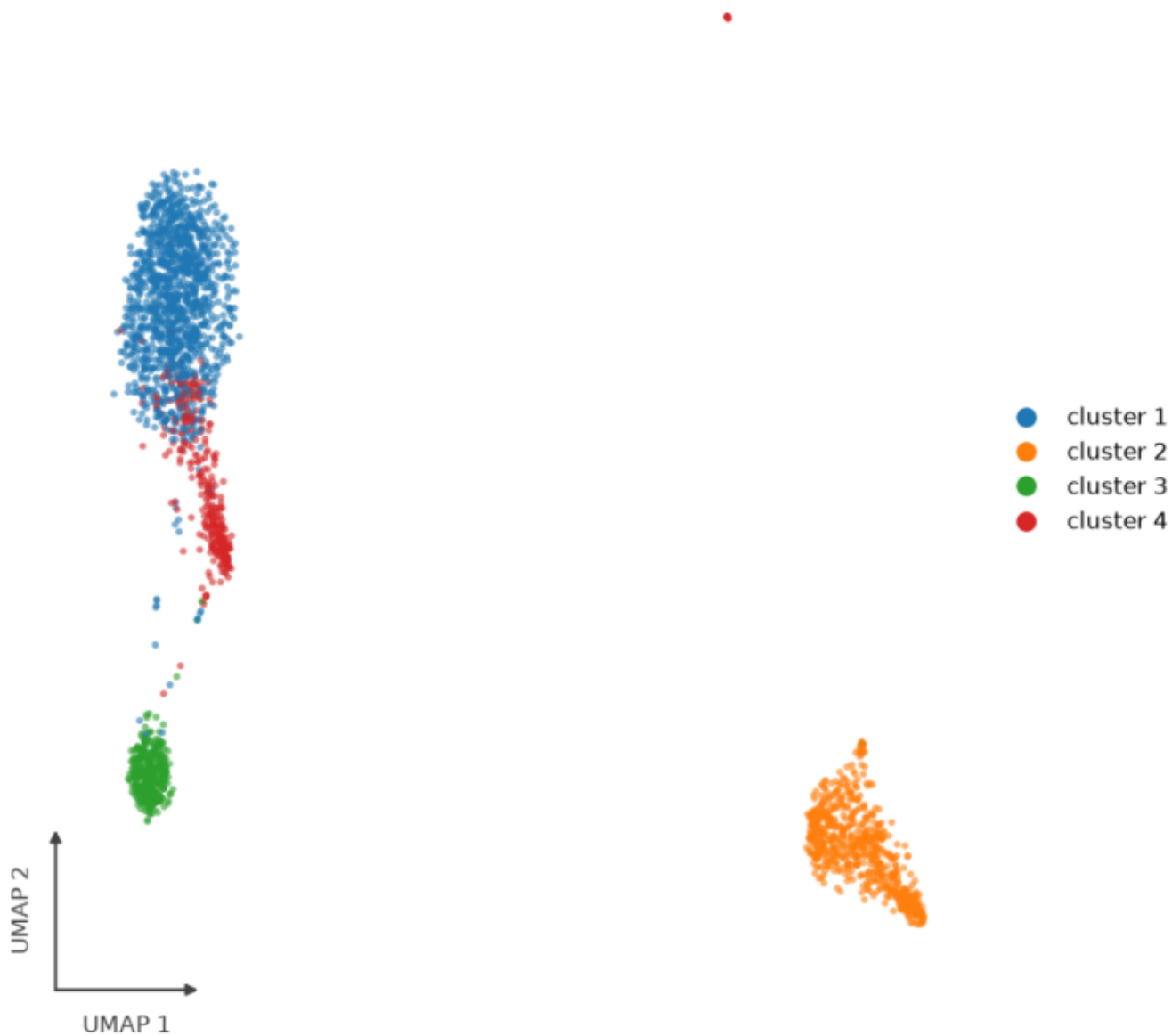
pbmc: diffusion_maps

n=2,700, colored by data-derived clusters (k-means on pca, k=4), showing dims 1-2 of 10



pbmc: umap

n=2,700, colored by data-derived clusters (k-means on pca,
k=4)

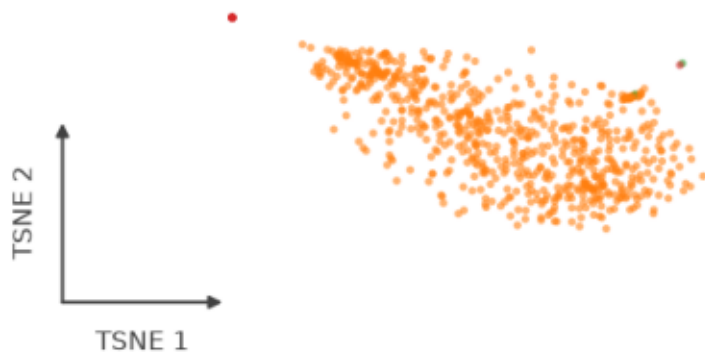


pbmc: tsne

n=2,700, colored by data-derived clusters (k-means on pca,
k=4)



- cluster 1
- cluster 2
- cluster 3
- cluster 4



Findings and interpretation

What each method produced

All five methods in the plan ran successfully on all 2,700 cells, using the same preprocessing: per-cell total-count normalization, log1p, and the 2,000 highest-variance genes. No method needed a retry or a fallback, and no metric was subsampled. The seed is 0 throughout.

PCA (50 components) captured 0.2904 of the total variance of the 2,000-gene matrix. The first component alone accounts for 0.1044, the second 0.0354 and the third 0.0226, after which the scree plot flattens quickly into a long tail of small, similar-sized components. 90% of the variance was not reached within the 200-component diagnostic fit (reported as >200). So most of the variance in the log-expression data is spread thinly across many directions, while a few leading components carry a clearly larger share. This fits sparse single-cell counts, where much of the per-gene variance is sampling noise.

Trustworthiness (10 nearest neighbors) was 0.9344 for PCA (50 dimensions), 0.9199 for Kernel PCA with an RBF kernel (50 dimensions), 0.8466 for Diffusion Maps (10 dimensions), 0.8023 for t-SNE (2 dimensions) and 0.7740 for UMAP (2 dimensions). These are not a head-to-head ranking. The embeddings have different numbers of dimensions, and a 50-dimensional embedding has far more room to keep neighborhoods intact than a 2-dimensional one. The fair comparisons are PCA against Kernel PCA (both 50 dimensions, similar scores) and UMAP against t-SNE (both 2 dimensions, t-SNE slightly higher on this measure). The leading Diffusion Maps eigenvalues were 0.1069 and 0.0342, with a similar steep drop after the first as in the PCA spectrum.

Findings and interpretation (cont.)

What the figures show

Every figure shows the same broad pattern: one group of cells (cluster 2 in the coloring) sits clearly apart from the rest. It is separated along the first component in PCA and Kernel PCA, along the first diffusion coordinate in Diffusion Maps, and it forms its own island in both UMAP and t-SNE. Because this split shows up in the linear methods, in both nonlinear general-purpose methods, and in two visualization methods with different objectives, it is the most robust structural observation in this analysis.

The remaining cells are not one uniform mass. In PCA, Kernel PCA and Diffusion Maps they spread along an elongated direction, and in PCA and Kernel PCA a smaller, more compact group sits just below that main body. UMAP and t-SNE both show that compact group as a separate island as well. Since it appears in both visualization methods and in the linear PCA view, it is likely a real feature of the preprocessed data, not a layout artifact. At $k=4$ this group is captured on its own (cluster 3, see below); at $k=3$ it was not, it stayed merged into the main body.

Both UMAP and t-SNE also place a very small number of cells as an isolated speck far from everything else. How many cells that is, and what they are, can't be read off a 2D visualization. Given that DATA_DESCRIPTION.md says no quality-control filtering was applied, a small set of unusual or low-quality cells is one possible explanation, but this is not tested here.

Findings and interpretation (cont.)

Data-derived clusters

The figures are colored by k-means on the 50-component PCA embedding, revised twice. The default scan over $k = 2$ to 10 picked $k = 2$, the lower bound, but the figures clearly showed more structure than two colors. Raising the floor to 3 picked $k = 3$ (silhouette 0.304), but that still merged the compact group described above into the main body, so $k=3$'s apparent fit to "more structure than 2 colors" was itself misleading: it just moved the coarse split, it didn't resolve the group visible in the figures. Rather than raise k again on the same basis (silhouette alone, or trusting how many islands UMAP happens to draw), the compact group was checked directly: a density-based grouping of the UMAP layout (used only to name a candidate set of points, not as a clustering result) isolates it as roughly 351 cells, and 345 of those land in one k-means cluster on the PCA embedding at $k=4$, versus 349 of 351 being absorbed into the main-body cluster at $k=3$. That confirms the group is separable in the general-purpose embedding itself, not just an artifact of UMAP's layout, so the floor was raised to 4 and the scan over $k = 4$ to 10 picked $k = 4$ (silhouette 0.289, barely below $k=3$'s 0.304). Silhouette then fell steadily as k grew further ($k=7$, at 0.206, is the only small uptick, over $k=6$ at 0.204), so the data do not favor any particular larger number of groups. The resulting cluster sizes are 1,330, 685, 348 and 337 cells. Cluster 3 (348 cells) is the compact group; cluster 4 (337 cells) instead splits the elongated main body along its own gradient (the long arm in the Diffusion Maps and Kernel PCA plots), the same behavior cluster 3 showed at $k=3$, now one level higher. These clusters are a coloring aid only. They are not validated groups, and their sizes should not be read as proportions of any biological population.

How the methods relate

PCA, Kernel PCA and Diffusion Maps are the general-purpose embeddings: they are the ones whose coordinates could reasonably feed further quantitative work, and PCA is the one the clustering was built on. They broadly agree with each other on the main split. UMAP and t-SNE are only pictures. They were included together so that no structural claim rests on a single visualization layout, and they agree on the main separated group, the compact second island and the small isolated speck. Neither was used for any computation beyond its own trustworthiness score.

Findings and interpretation (cont.)

What cannot be concluded

This dataset has no labels, so none of the groups can be named, and nothing here shows that they correspond to cell types. The distances between islands and the sizes and shapes of islands in UMAP and t-SNE carry no quantitative meaning. The data were not quality-filtered, and the pipeline has no QC step, so low-quality cells or doublets may influence both the leading components and the small outlying group. Feature selection ranked genes by raw variance rather than mean-adjusted dispersion, which can favor highly expressed genes; a different gene selection could change the finer structure. Finally, each method was run with one seed; the UMAP/t-SNE agreement is the only cross-check on the visual structure.

Limitations

No ground-truth labels available for this dataset; embedding quality assessed only via label-free trustworthiness, no supervised sanity check possible.

The cluster colors on unlabeled plots come from k-means ($k=4$) on the pca embedding, with k picked by silhouette rather than validated against any ground truth. Different k , a different algorithm, or a different source embedding would color the same points differently; treat the colors as a visual aid, not as groups that are known to exist.

No adversarial critique/revision pass was performed for this run (critic subagent not yet built).